

Back of thigh (hamstring) & POPliteal fossa

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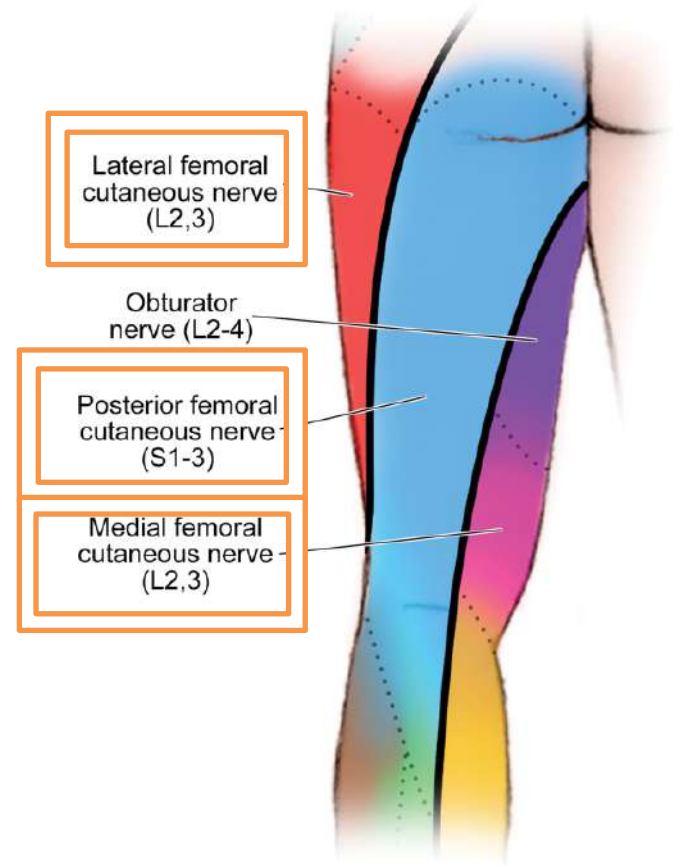
2025





Cutaneous nerve supply

- 1- The medial cutaneous nerve of the thigh on the medial side (branch of femoral nerve)
- 2- Posterior cutaneous nerve of the thigh on the middle S1,2,3
- 3- Lateral cutaneous nerve of the thigh on the lateral side L2,3



Cutaneous nerve	Origin	Area of skin supply in thigh
.Posterior cutaneous nerve of the thigh.	Branch of sacral plexus (S1,2,3).	Supply back of the thigh, roof of popliteal fossa & upper part of the back of calf.

❖ Content :

❖ Muscles:

1-Biceps femoris.

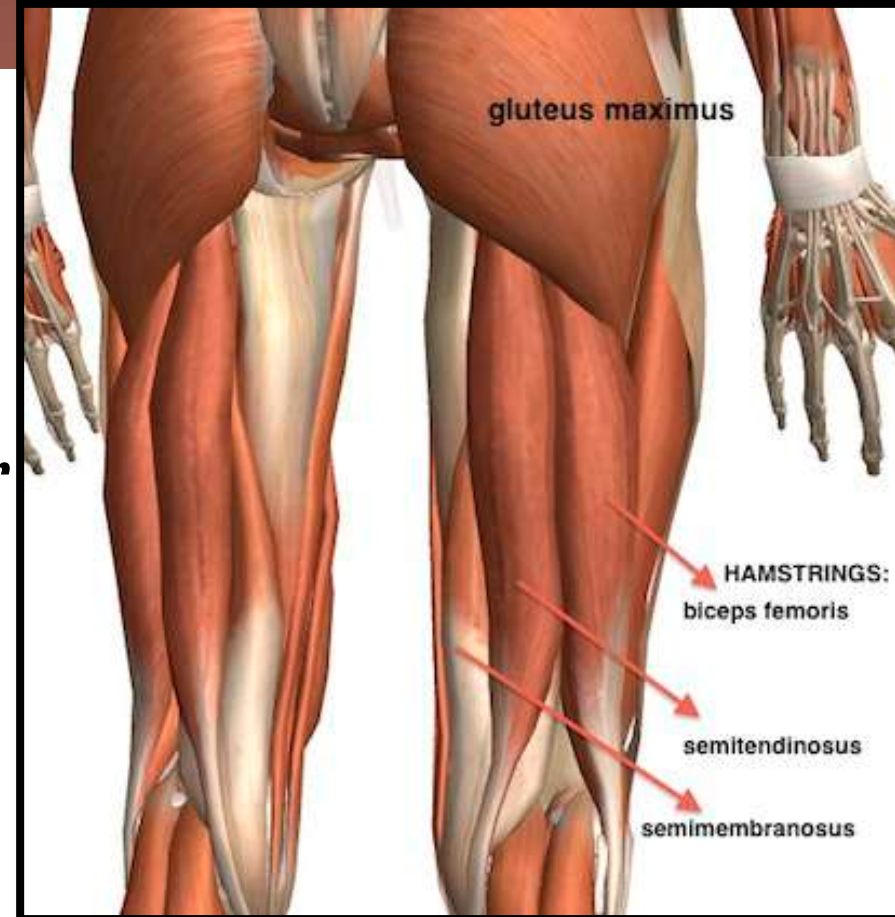
2-Semitendinosus.

3-Semimembranosus.

4-Hamstring portion of adductor magnus.

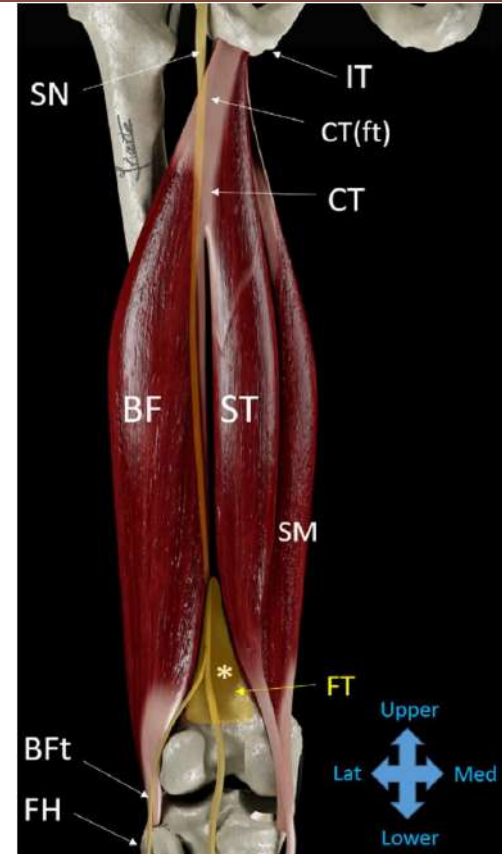
❖ Blood supply: branches of Profunda femoris artery.

❖ Nerve supply: Sciatic nerve.



❖ Muscles of the back of thigh:

- ❖ They form mass on the back of the thigh.
- ❖ **Ham**=back of thigh / **strings**= tendons = **hamstring muscles**.
- ❖ They have a long course from origin to insertion.
- ❖ All innervated by sciatic nerve.
- ❖ All arise from **Ischeal tuberosity** Except short head of **biceps**.
- ❖ All are inserted into bones of the legs Except **hamstring portion of adductor magnus(in femur)**
- ❖ They are **extensors** for hip joint & **flexors** of the knee joint (**important in walking**).



ذات الرأسين الـ **Biceps femoris m** فخذية (most lat. Muscle of hamstrings)

O: **1) Lateral head: Short head (gluteal portion)**

; from linea aspera, & upper $\frac{1}{2}$ of lateral supracondylar ridge of shaft of femur.

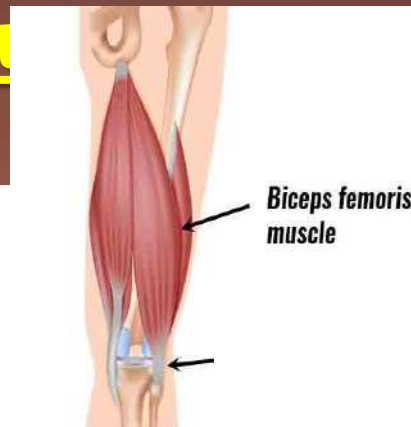
2) Medial head: Long head (hamstring portion); from ischeal tuberosity.

I : Two heads unite above knee joint—by **common tendon** inserted into **head of Fibula**.

N : **Long head:** Tibial part of sciatic nerve.

Short head: Common peroneal part of sciatic n.

-Double nerve supply!!



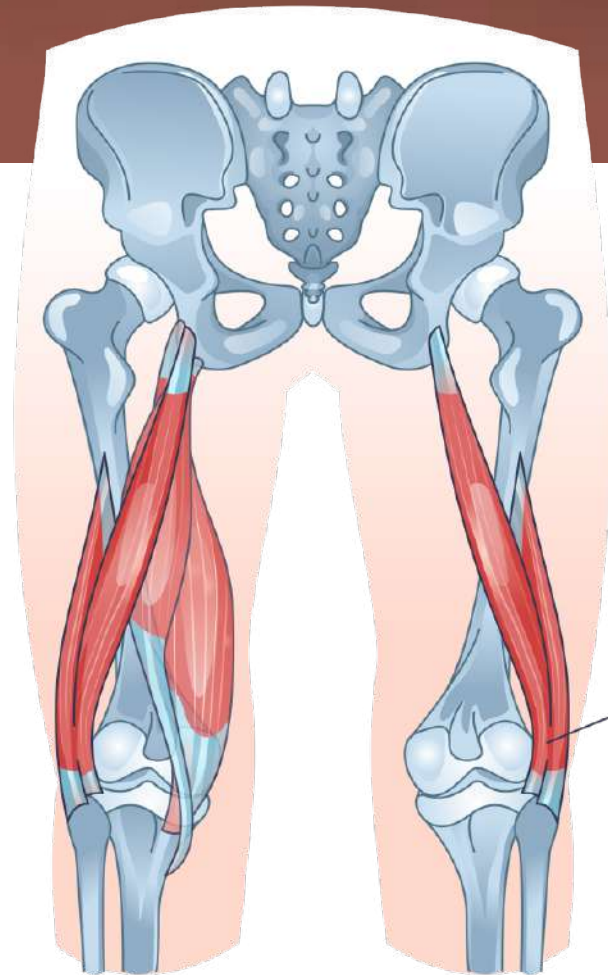
A: hip joint: long head; Extends thigh at hip joint.

B: knee joint: flexion & lateral rotation of leg at the knee joint **(the only lateral rotator of the leg at the knee joint)**





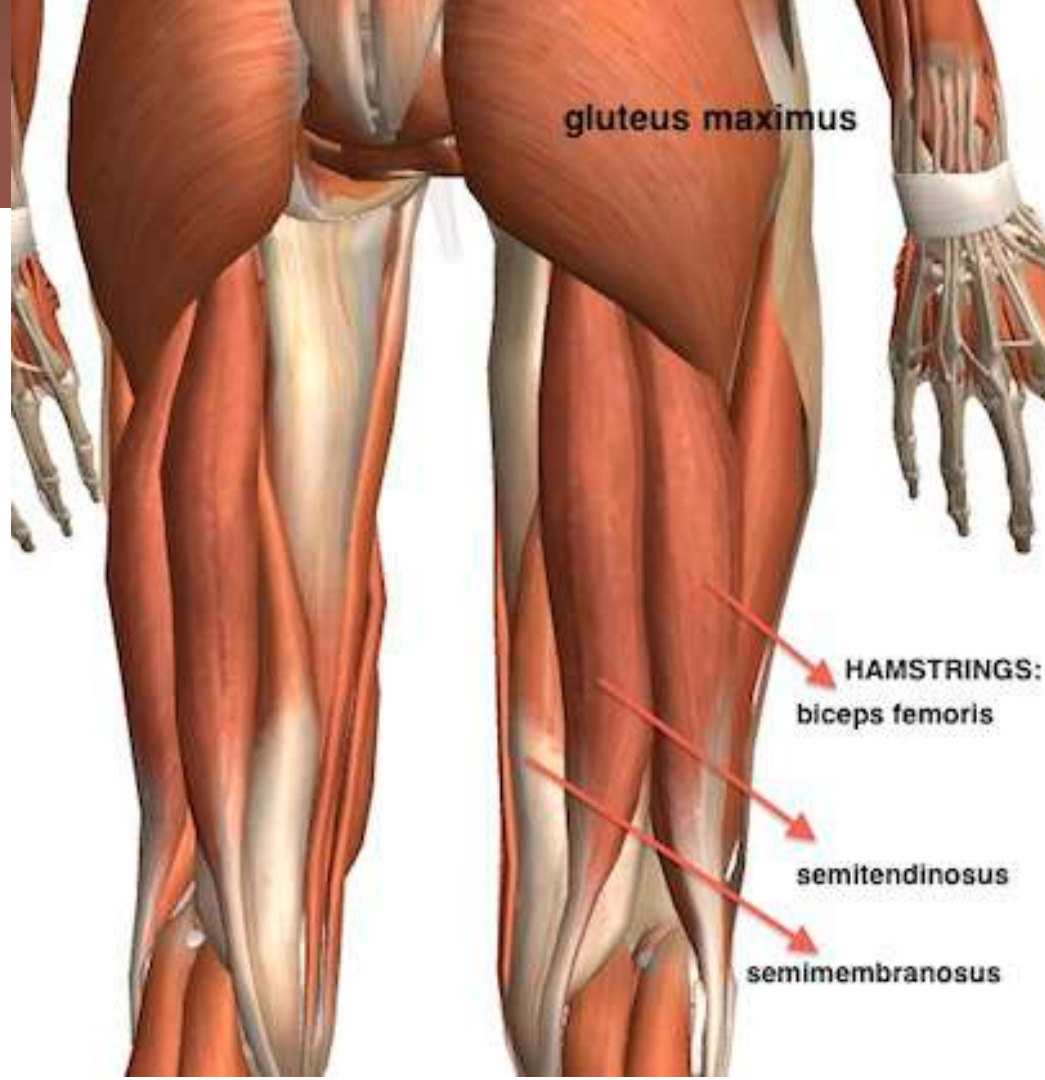
- Role of biceps femoris in “locking” of knee joint:
- Locking of knee joint means: **lateral rotation** of the leg **at the end of extension of the knee**, which is performed by quadriceps femoris.
- It is ***only lateral rotator*** of the knee joint.



**BICEPS FEMORIS
MUSCLE**

On the cadaver
from lateral to
medial: على الجثة

- 1) Biceps femoris.
- 2) Semitendinosus.
- 3) Semimembranosus.



2-Semitendinosus نصف الوترية

(Fusiform belly & long cord-like tendon)

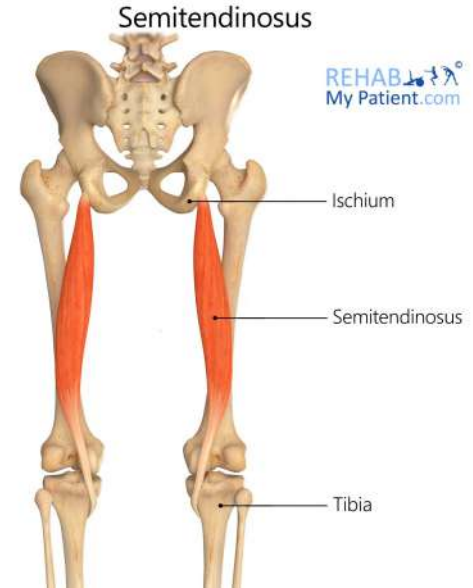
O: from ischeal tuberosity.

I : By a long tendon into the **upper part of the medial surface of shaft of tibia (SGS).**

N : **Tibial portion** of sciatic nerve.

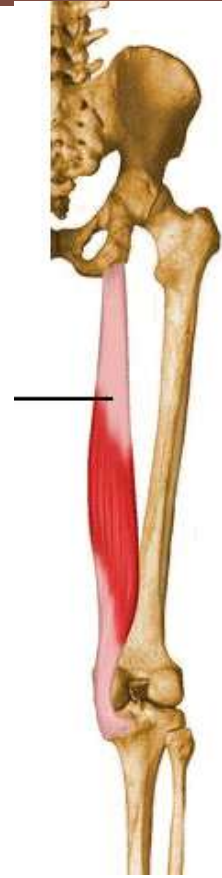
A : **hip joint :** *extension* of the thigh at hip joint.

knee joint: flexion & medial rotation of the leg at the knee joint.

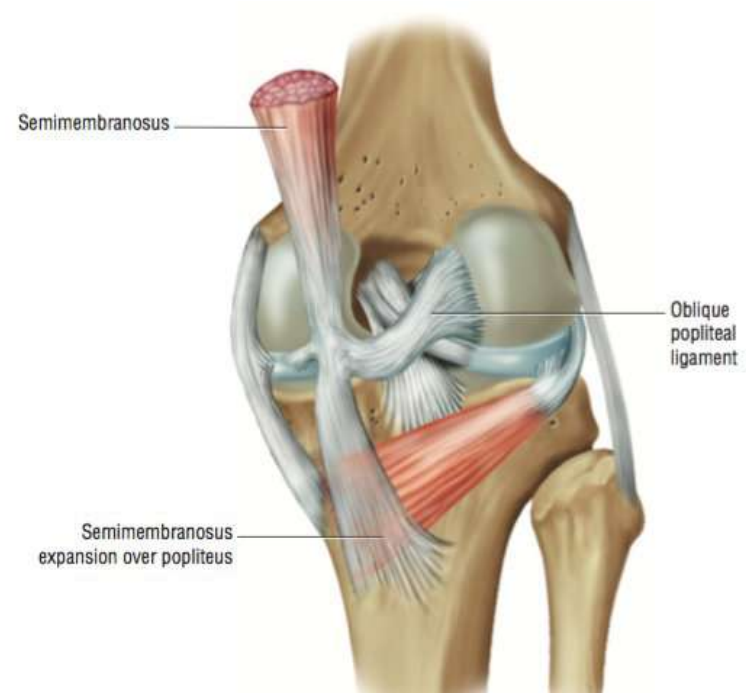


3-Semimembranosus نصف الغشائية

- O:** Ischeal tuberosity.
- I :** Postero-medial surface of the **medial condyle** of the tibia.
- N :** **Tibial portion** of sciatic nerve.
- A :** **hip joint:** extension of the thigh at hip joint.
knee joint: flexion & medial rotation of the leg at the knee joint.



- **Oblique popliteal ligament**: fibrous expansion from **semimembranosus** to the capsule on the back of the knee joint, which reinforces the capsule at this site.



4- Hamstring portion of Adductor magnus

❖ It forms the thickest part of the muscle.

O : Ischial tuberosity.

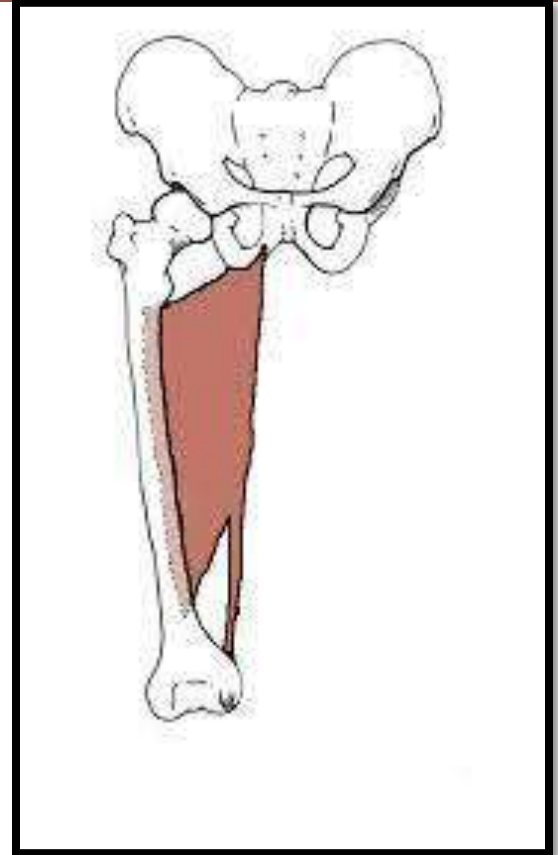
I : to the **adductor tubercle** on the medial epicondyle of femur..

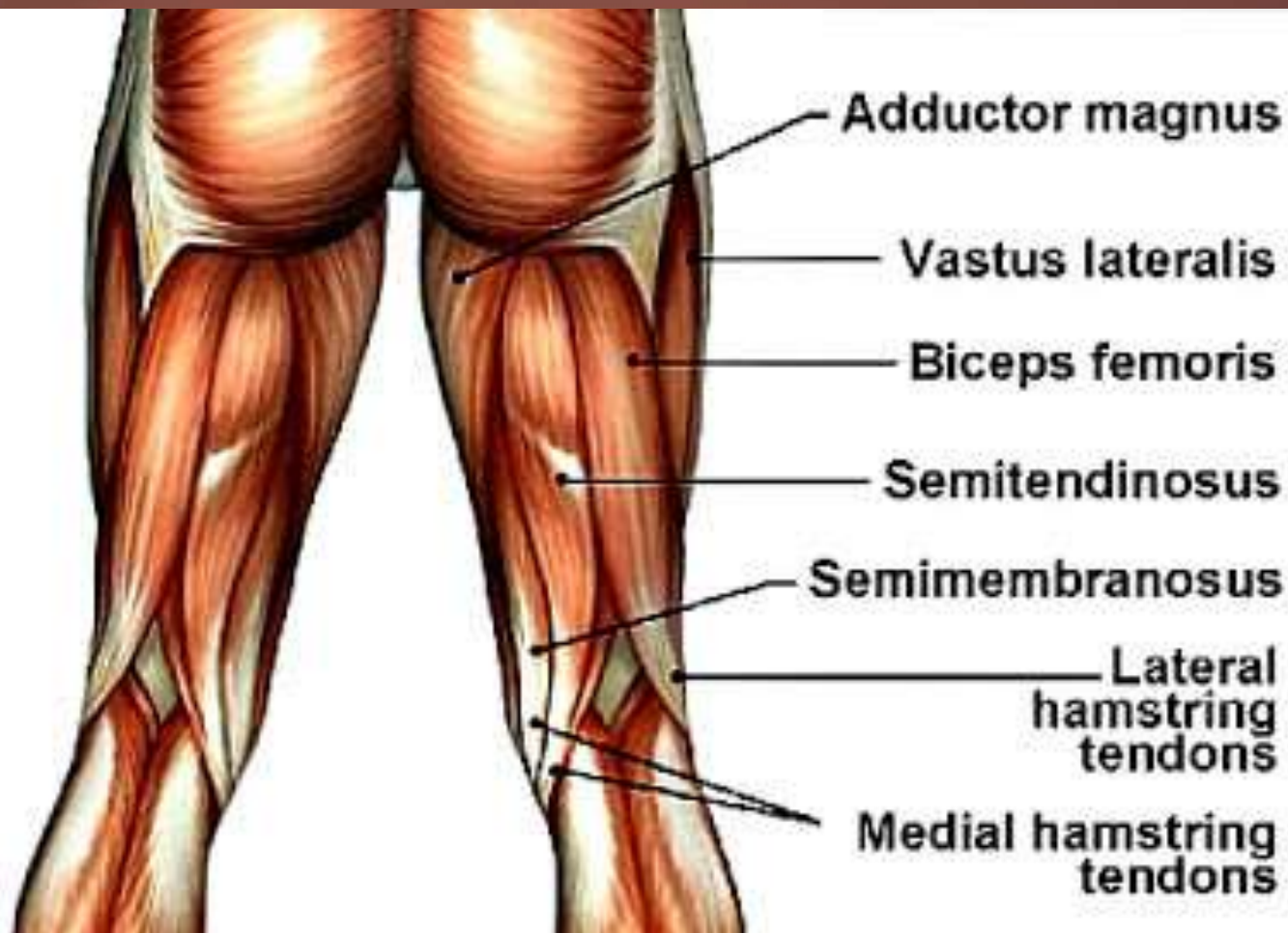
N : Tibial portion of sciatic nerve.

A: Extension of thigh at hip joint.



It acts **only** on the hip joint and not the knee joint.





- ❖ All the hamstringing muscles have origin from ischeal tuberosity EXCEPT short head of biceps femoris.
- ❖ All the hamstringing muscles are supplied by tibial part of sciatic nerve EXCEPT short head of biceps (by common peroneal) part. مهم
- ❖ All hamstringing muscles act on both joints(hip &knee) Except hamstringing part of adductor magnus, acts only on hip joint & short head of biceps acts on knee joint.
- ❖ **SGS:** sartorius, gracilis, semitendinosus.



muscle	hip	knee
1-Biceps femoris	Extension	Flexion& lateral rotation
2-Semi-T	Extension	Flexion & medial rotation
3-Semi-M	Extension	Flexion & medial rotation
4-hamstering portion of add.magnus	Extension	-----

- 1) Flexors of the thigh: Pectineus, iliacus, psoas, sartorius, rectus femoris, gracilis.
- 2) Extensors of the thigh: hamstring portion of adductor magnus.& **gluteus maximus**, long head of biceps femoris, Semimembranosus, Semitendinosus.
- 3) Lateral rotators of the thigh: Obturator externus, sartorius, adductor longus, brevis & magnus. , **Piriformis, Gemellus Superior, Gemellus inferior, Obturator internus, Quadratus Femoris. Gluteus maximus.**
- 4) Medial rotators of the thigh: iliacus, psoas.+ **anterior fibers of gluteus medius. & minimus.**
- 5) Adductors of the thigh: Pectineus, adductor longus, adductor brevis, adductor magnus & gracilis.
- 6) Abductor of the thigh: Sartorius, **Gluteus medius and gluteus minimus**
- 7) Flexion & medial rotators of the knee: Sartorius, Gracilis, **biceps femoris, Semitendinosus, Semimembranosus,**
- 8) Flexion & lateral rotator of the knee: Biceps femoris.
- 8) Extensors of the knee: Quadriceps femoris.

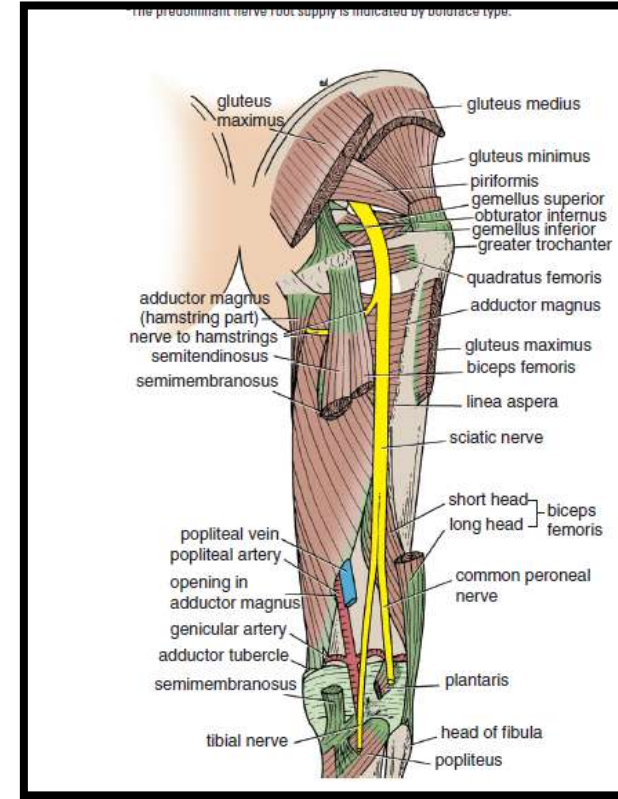
- ❑ The only lateral rotator of the knee joint is biceps femoris muscle.
- ❑ **Medial rotators of the knee joint** :Sartorius, Gracilis, Semi.T & Semi.M.(**SGSS**).

Blood supply of posterior compartment of thigh

- ❖ Four **perforating branches** of profunda femoris artery.
- ❖ **Profunda femoris vein** drains the greater part of blood from the compartment.

Nerve supply: Sciatic nerve (عرق النسا)

- ❖ Branch of **sacral plexus** (L4,5- S1,2,3).
- ❖ **Course:** leaves the gluteal region as it descends in the midline of the thigh.
- ❖ It lies on the post.aspect of **adductor magnus**.
- ❖ **End:** In the lower third of thigh, it divides into tibial(**Medial**) & common peroneal(**Lateral**) nerves.
- ❖ Occasionally, early division (pelvis, gluteal region, upper thigh) could happen.



❖ Branches

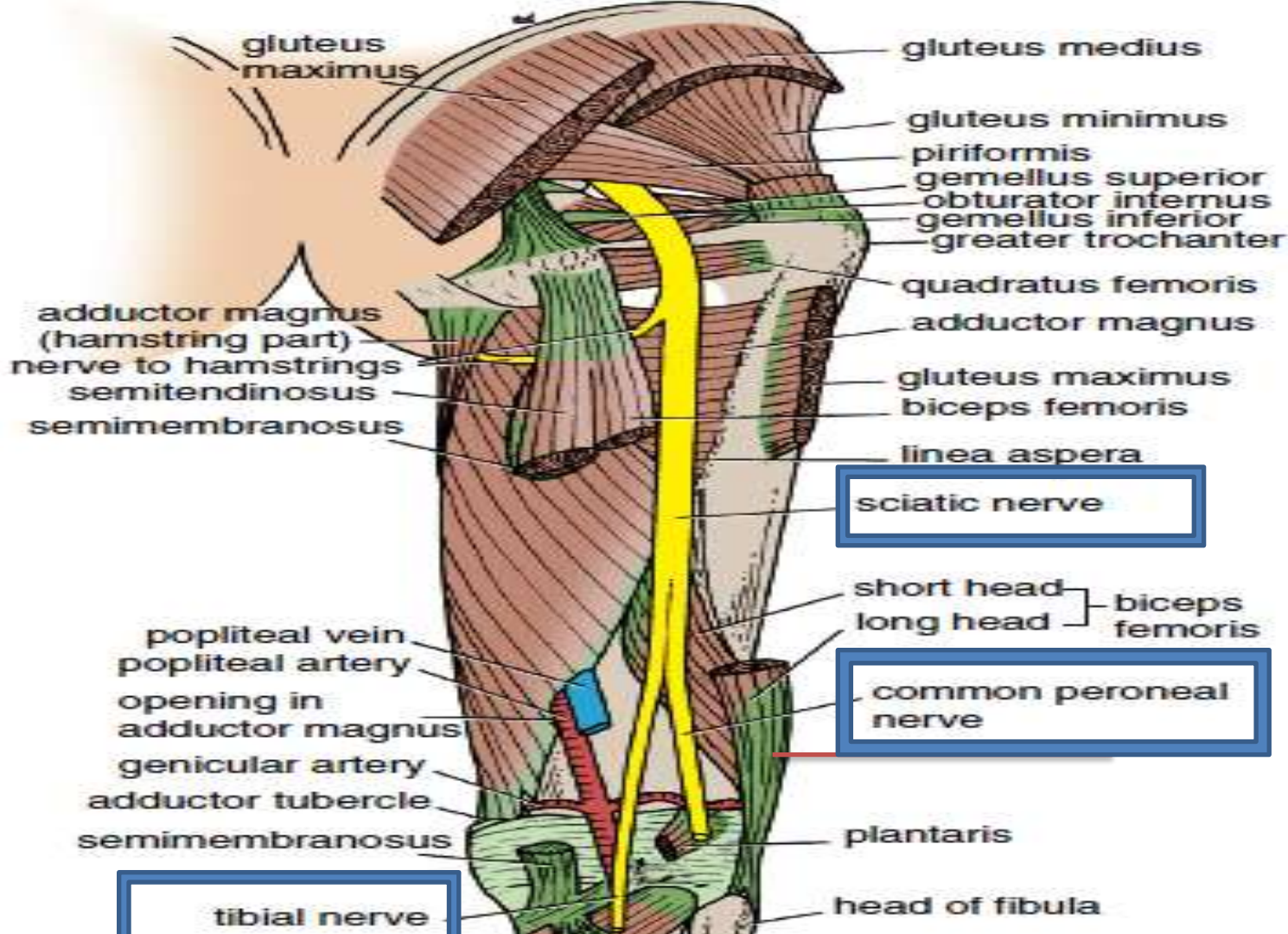
1- Tibial nerve: *Medial* Terminal branch of the sciatic nerve, enters the popliteal fossa.

2-Common peroneal nerves: *Lateral* Terminal branch of the sciatic nerve. Enters the popliteal fossa on the lateral side of tibial nerve.

3-Muscular branches:

A- From Tibial component of sciatic nerve: to the *long head of biceps femoris, semitendinosus, semimembranosus, hamstring part of adductor magnus.*

B- From Common peroneal component of sciatic nerve:
To **short head of biceps femoris** muscle.



Popliteal fossa

IMPORTANT

- ❑ After dissection, is a diamond –shaped, important intermuscular space.
- ❑ **Site:** situated at the back of the lower part of the thigh, knee, and upper part of the leg.
- ❑ formed between muscles in the post. Compartments in the thigh and leg.
- ❑ **It has :** boundaries, floor, roof& content.



The fossa is most prominent
when the knee joint is flexed.



Popliteal fossa

❖ **Boundaries** (Laterally, medially, anterior wall, posterior wall).

❖ **Importance**

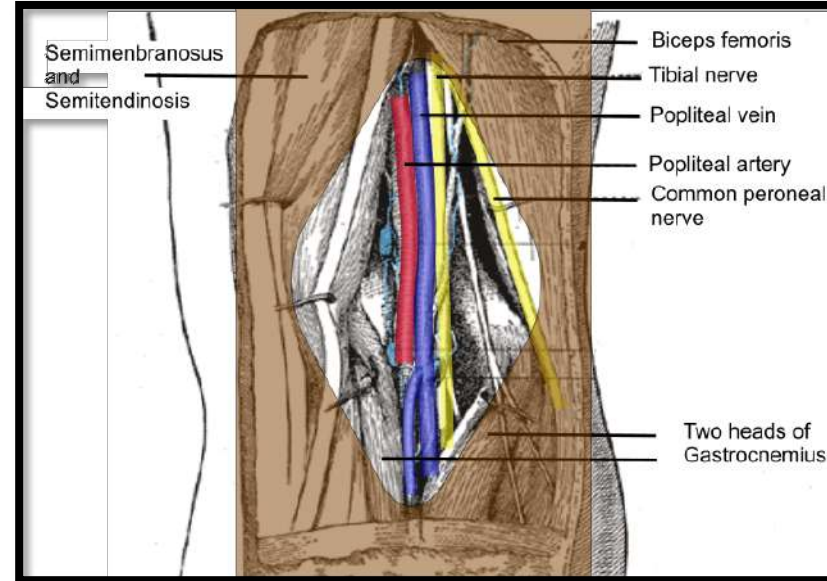
❖ **Content**



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□ Importance:

-Important area of transition between the thigh and leg & is a major route by which structures pass from one region to another.

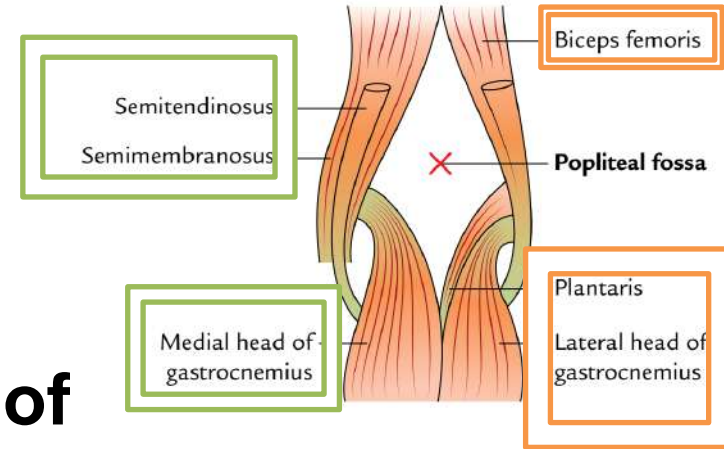


Boundaries

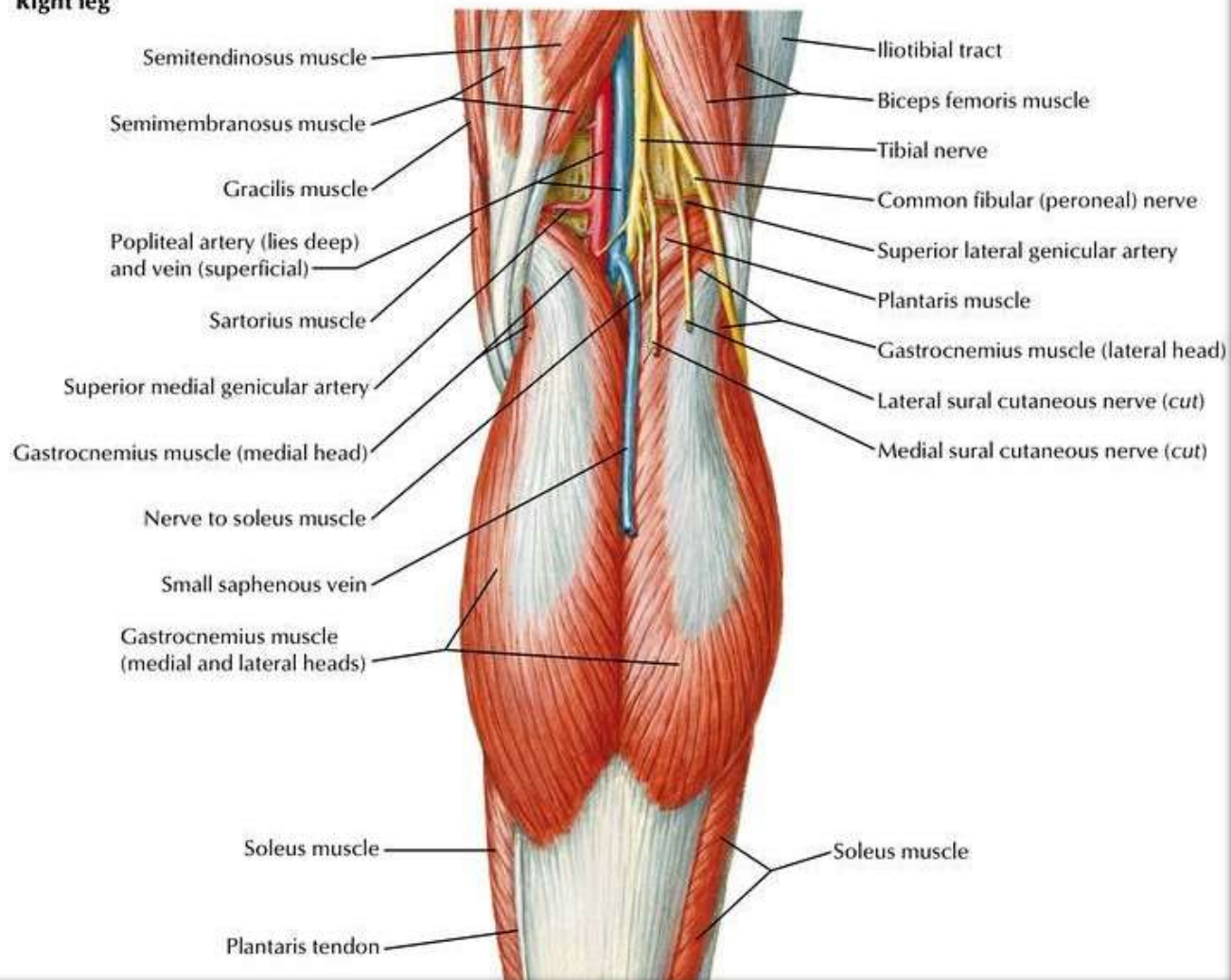
❑ **Laterally:** Above: Biceps femoris.
Below: Lateral head of gastrocnemius & plantaris.

❑ **Medially:** Above:
Semimembranosus & Semitendinosus .

Below: Medial head of gastrocnemius.



Right leg



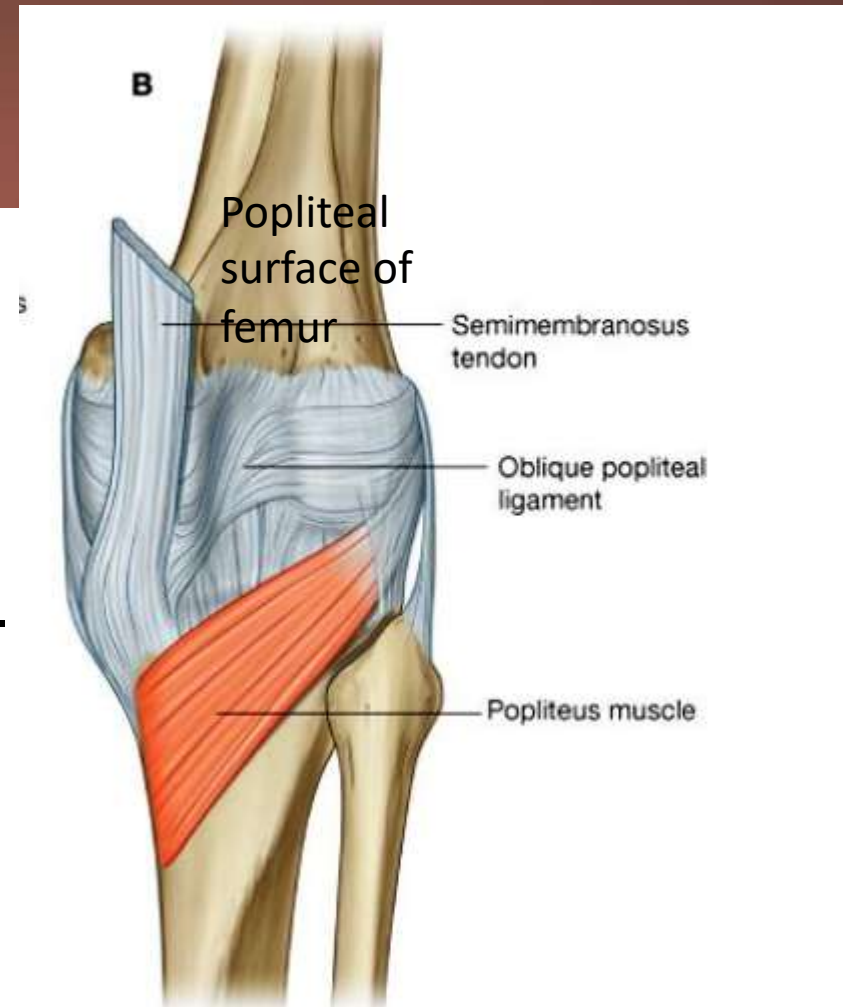
Plantaris



□ Anterior wall (Floor)

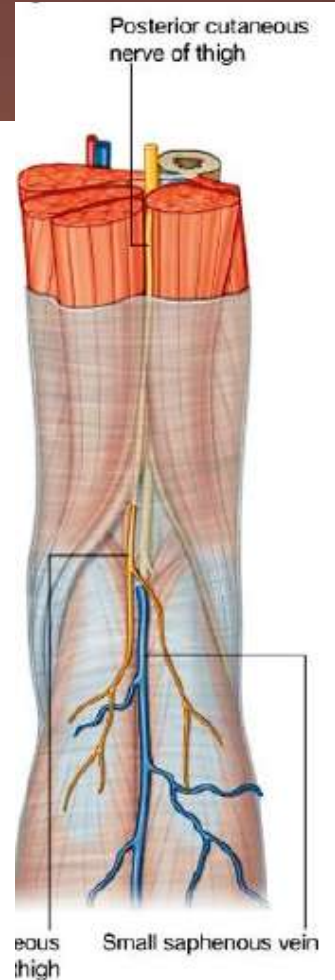
Formed by : (PPP)

- 1-**P**opliteal surface of **f**emur.
- 2-**P**osterior **l**igament of knee joint.
- 3-**P**opliteus **m**uscle.



□ Posterior wall (Roof):

- Skin, superficial fascia & deep fascia.
 - **Popliteal fascia** is strong sheet of deep fascia continuous superiorly with fascia Lata & inferiorly with deep fascia of leg.
- Most important structure in superficial
- Fascia is the upper part of small saphenous vein ending In popliteal vein.



□ Content:

A: Popliteal Artery

V: Popliteal Vein

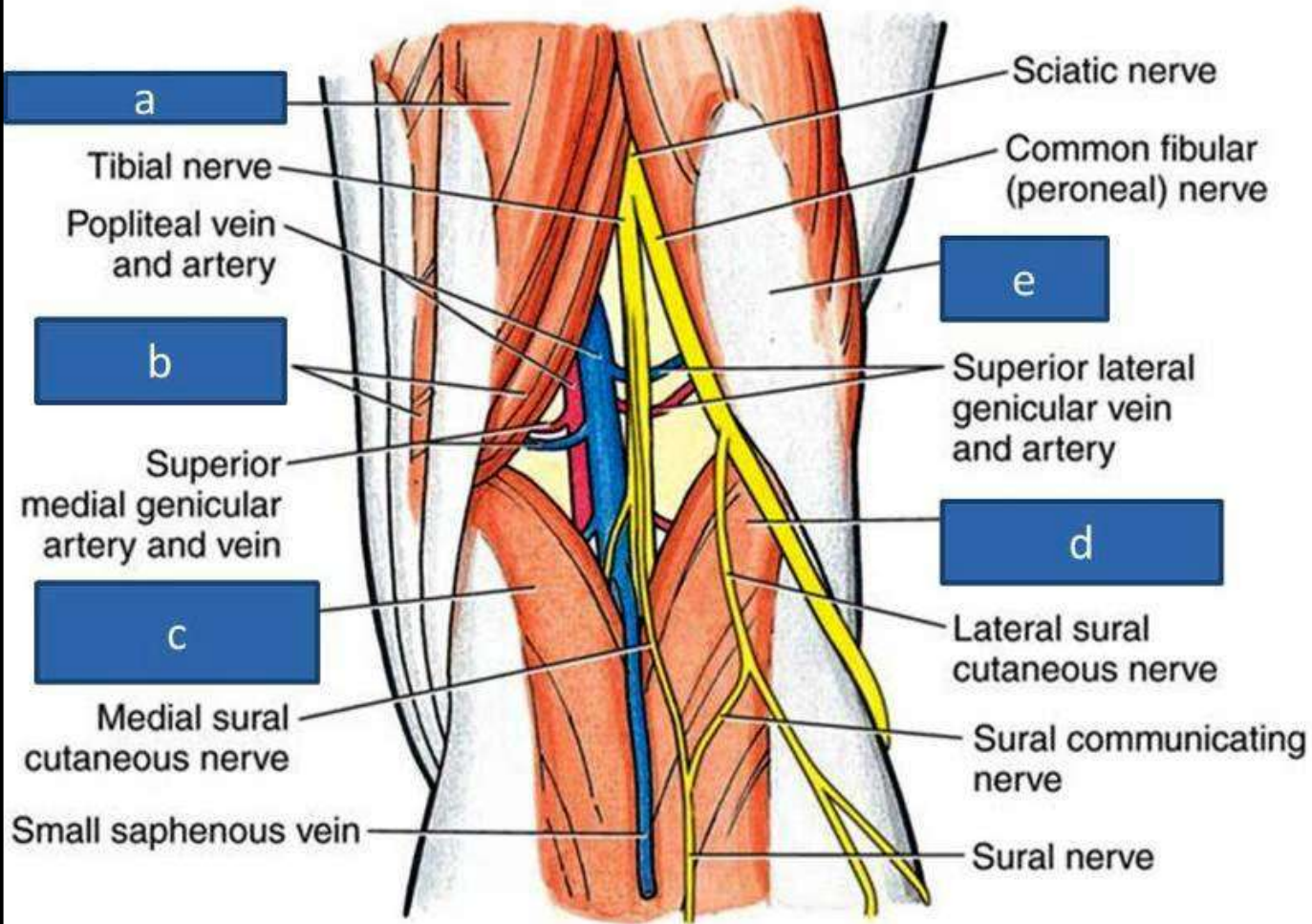
N: Tibial Nerve

N: Common peroneal Nerve.

4 structures from Medial to lateral
(**A****V****N****N**)

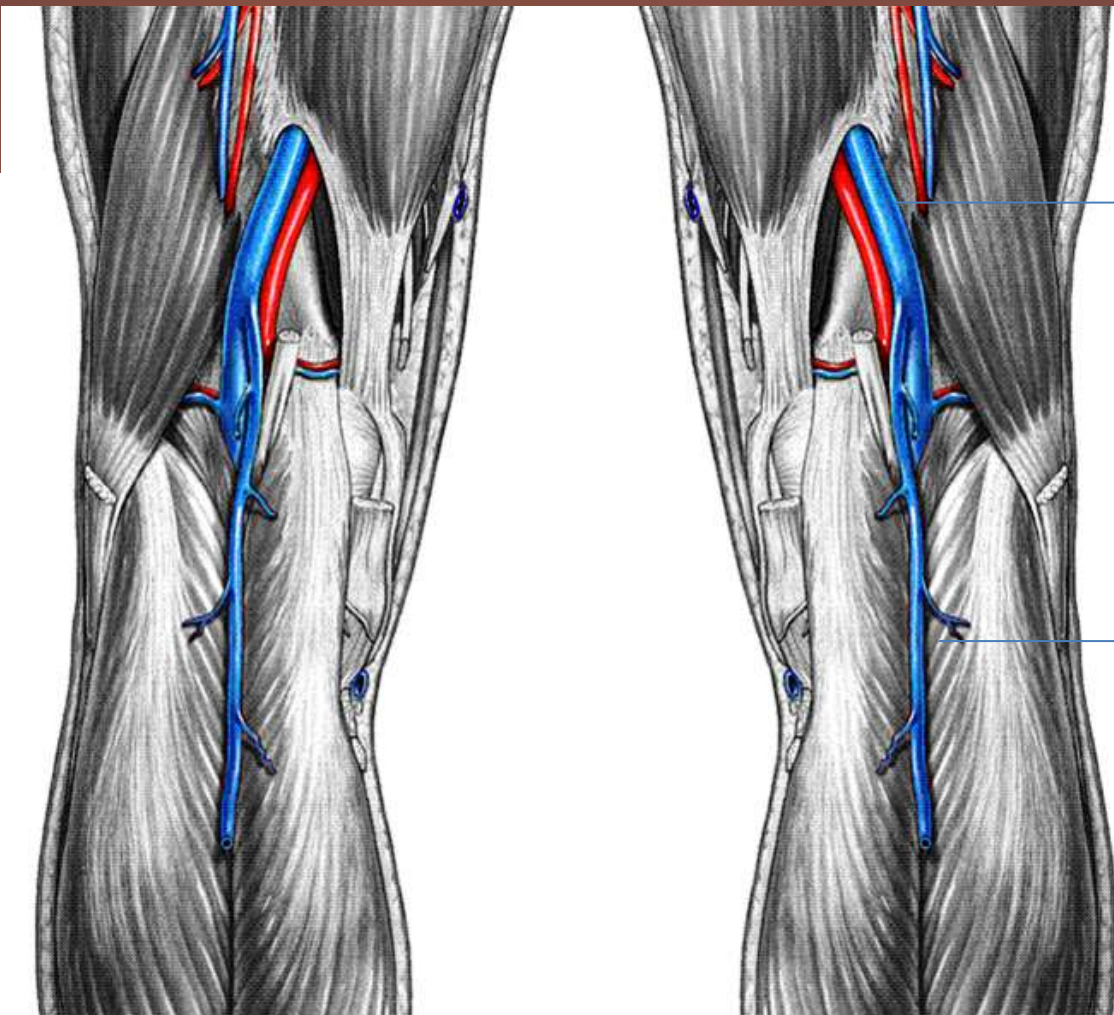
- + termination of Small saphenous vein
- + Lymph vessels & popliteal lymph nodes (superficial & deep).
- + Genicular branch of obturator nerve.
- + Posterior cutaneous nerve of the thigh
- + Connective tissue & popliteal fat.

Popliteal artery is the deepest structure in the fossa, lying on popliteal surface of femur. مهم



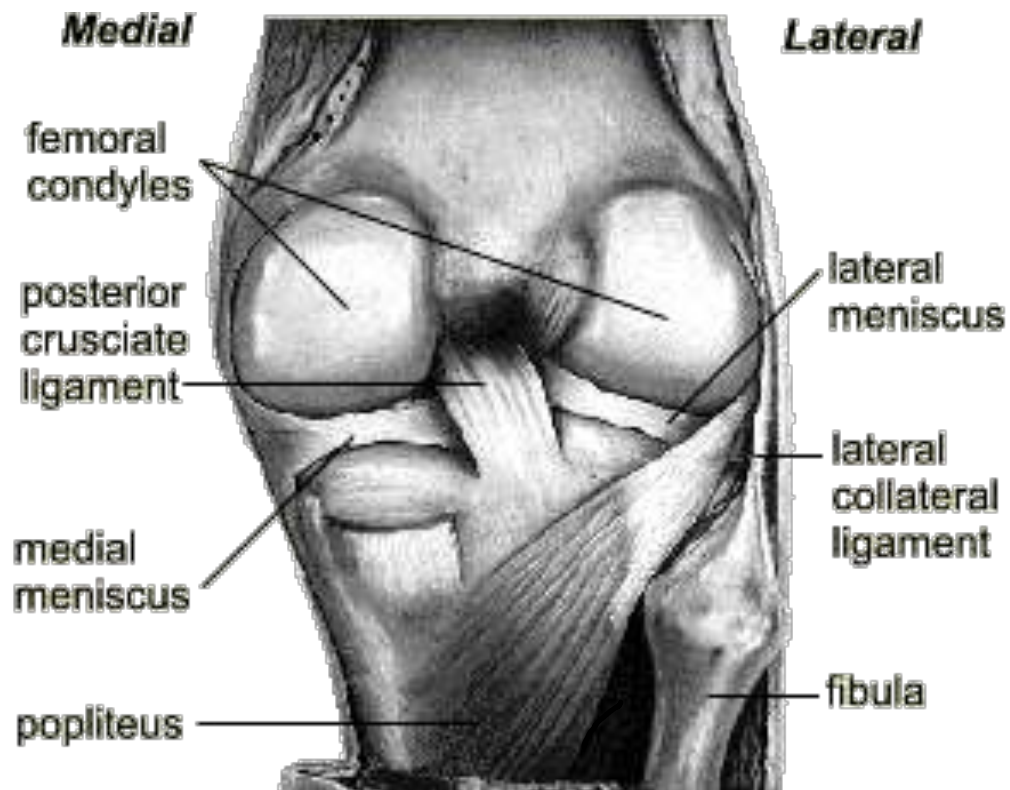
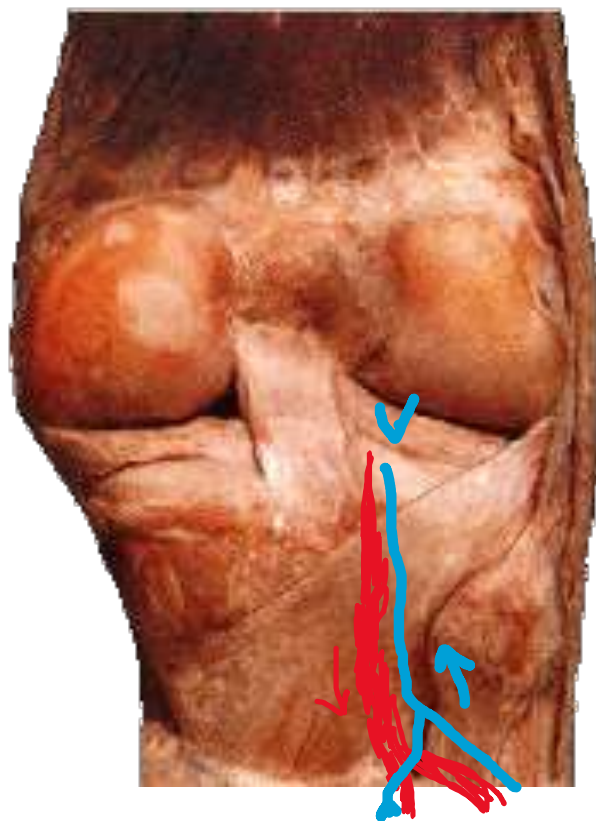
C Posterior view

(AVNN)
From
medial
to lateral



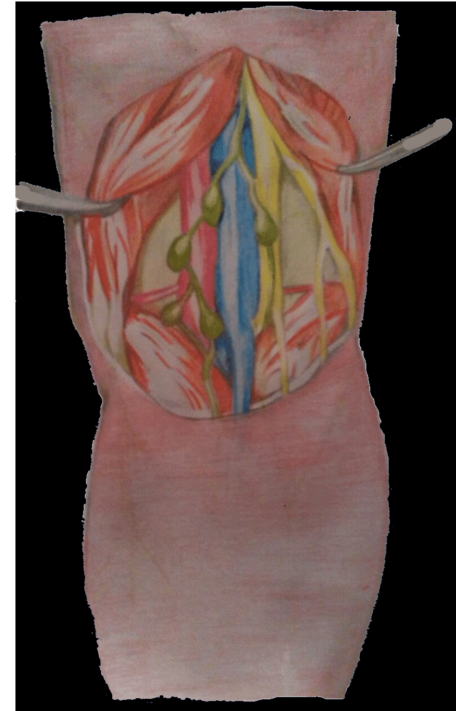
Popliteal
vein

Small
saphenous
vein



Popliteal lymph nodes

- About 6 in number embedded in popliteal fossa.
- They receive superficial lymph vessels from **lateral side of the foot& leg**, that accompany the small saphenous vein into the popliteal fossa.
- They also receive lymph from the **knee joint& from deep lymph vessels** accompanying the anterior and posterior tibial arteries.



❑ Popliteal artery

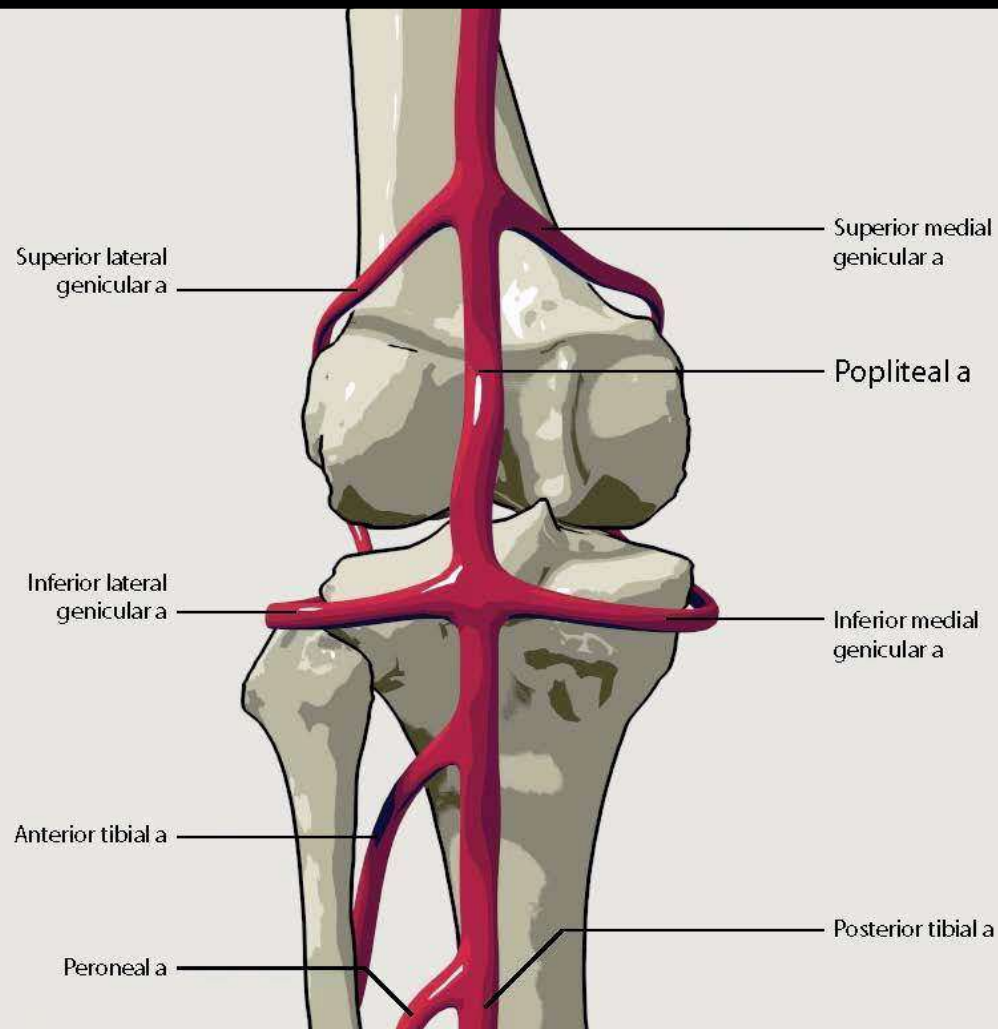
- ❖ **Beginning:** enters popliteal fossa through adductor hiatus, as a continuation of **femoral artery**. It is **deeply** placed in the fossa.
- ❖ **End:** at the level of lower border of the popliteus muscle by dividing into **anterior and posterior tibial arteries**.
- ❖ **Branches:**
 - 1-Muscular: to muscles
 - 2-Articular branches: to the knee

❖ Relations:

Anteriorly: popliteal surface of the femur, the knee joint and popliteus muscle.

Posteriorly: popliteal vein and tibial nerve, fascia and skin.

تذكري دائماً " وضع الجثة في prone position



Arterial anastomosis around the knee joint:

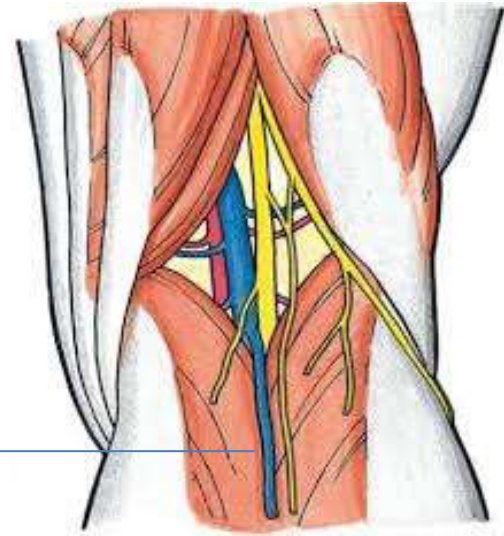
-Profuse anastomosis of small branches of femoral artery with muscular & articular branches of the popliteal artery & branches of anterior & posterior tibial arteries.

-Is necessary to compensate for the narrowing of the popliteal artery that occurs during extreme flexion of the knee.

Popliteal vein

What are venae comitantes???
متسلقات وريدية

- ❑ **Beginning** :formed by union of venae comitants of the **anterior& posterior tibial arteries** at the lower border of popliteus, on the **medial side** of popliteal artery.
- ❑ **Course**: As it ascends through the fossa, it crosses behind popliteal artery so that it comes **to lie on its lateral side**.
- ❑ It Passes through the opening in adductor magnus to become **femoral vein**.
- ❑ **Tributaries**:
 - 1-Veins correspond to branches of popliteal artery.
 - 2-*Small saphenous vein*: perforates the deep fascia passing between the two heads of gastrocnemius to end in **popliteal vein**.





Vinea comitants
forming popliteal
vein



(Formative assessment)

1-Which one of the following muscles passes through the greater sciatic foramen?

- A. Superior gemelus
- B. Obturator externus
- C. Obturator internus
- D. Piriformis

4-The adductor hiatus present in which of the following muscles?

- A. Adductor longus
- B. Adductor magnus
- C. Adductor brevis
- D. Adductor hallucis

6-Which of the following structures **does not** enter the femoral sheath ?

- A. Femoral nerve
- B. Femoral artery
- C. Femoral vein
- D. Lymph nodes



عندما يتحدث الناس عنك بسوء
وأنت تعلم أنك لم تخطئ في حق
أحد منهم تذكر أن تحمد الله الذي
أشغلهم بك ولم يشغلك بهم.